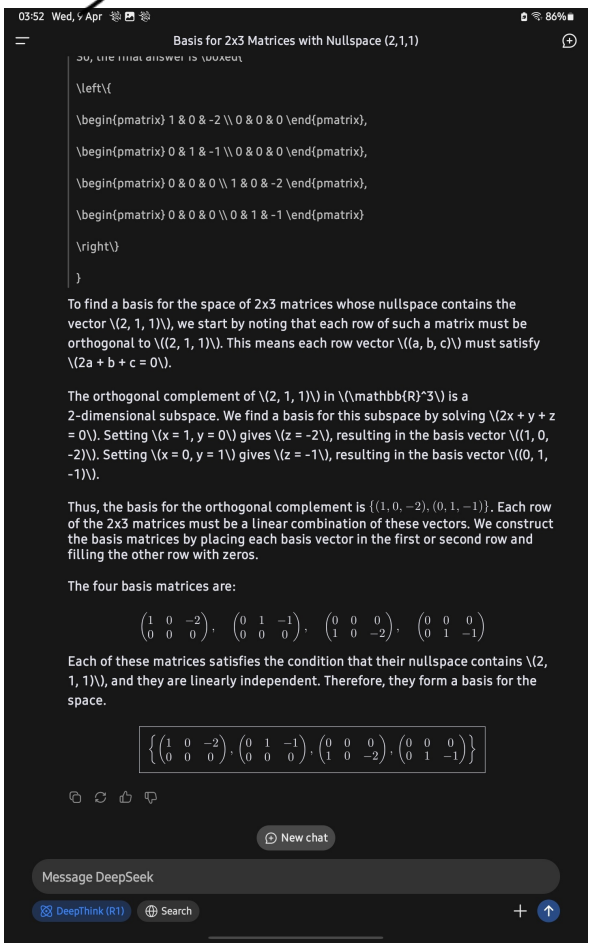
31) 2x3 → N(A) → 2, 1, 1



37) S → (a, b, c, d) → a + c + d = 0

a = −c − d = 0

∴ b, c, d free var.

a, b, c, d = (c − d, b, c, d) = b(0, 1, 0, 0), c(−1, 0, 1, 0)

+ d(−1, 0, 0, 1)

=> reorder for cleaner basis:

$(1, 0, -1, 0)$, $(0, 1, 0, 0)$, $(1, 0, 0, -1)$

$T \Rightarrow \quad a+b=0, \quad \therefore b=-a, \quad c=2d$

a, d free var.

$(a, b, c, d) = (a, -a, 2d, d) = (1, -1, 0, 0),$

$(0, 0, 2, 1)$

$\therefore$ basis for T $(1, -1, 0, 0), (0, 0, 2, 1)$

$S \cap T$:

we want all vectors that satisfy both:

$a+c+d=0$ — (i)

$a+b=0$ — (ii)

$c=2d$ — (iii)

} 3 eqn in R^4

Lets use (ii) & (iii) in (i),

$a+2d+d=0 \quad \therefore a=-3d \quad \therefore b=3d,$

$c=2d.$

any vector in this intersection:

a, b, c, d:- $(-3d, 3d, 2d, 2) = d(-3, 3, 2, 1)$

OR $(3, -3, 2, 1)$

Dim $S \cap T = 1$

(19) $(1, 1, 1)$

(23)

(20) $x - 2y + 3z = 0$

$x = 2y - 3z$

$(2, 1, 0), (-3, 0, 1)$

xy plane →

$x - 2y = 0$

$x = 2y$

$(2, 1, 0)$

16 b) all vectors whose component add to zero in R^4.

$x_1 + x_2 + x_3 + x_4 = 0$

$x_1 = -x_2 - x_3 - x_4$

$(-1, 1, 0, 0)$